

Technical Specifications, Environmental Durability & Lifecycle Analysis

Nordinfra Systems — US Market

Material & Coating Architecture

Our ground foundations are engineered from premium Grade 33 structural steel, hot-dip coated with a Zinc-Aluminum-Magnesium (ZAM®) alloy in a ZM115 coating class, in full compliance with ASTM A1046/A1046M (Standard Specification for Steel Sheet, Zinc-Aluminum-Magnesium Alloy-Coated by the Hot-Dip Process). Grade 33 provides a certified minimum yield strength of 33 ksi, and the ZM115 coating provides 1.15 oz/ft² (approximately 350 g/m²) of total surface protection — a coating weight comparable to or exceeding the ZM310 material used on our Swedish-market foundations. Independent published corrosion studies indicate Zinc-Aluminum-Magnesium coatings of this type outperform traditional G90 galvanized steel by a factor of roughly three to five in buried/soil environments, depending on soil conditions.

Note: the Small and Medium Adapter Plates are manufactured from separate A36 steel and hot-dip galvanized after fabrication (per ASTM A123/A153), rather than supplied as pre-coated ZM material — this ensures full coverage of drilled and countersunk features added after cutting.

Laser-Cut Edge Protection

Traditional zinc-galvanized steel leaves the raw substrate exposed at a laser-cut edge, creating a localized point of vulnerability. Zinc-Aluminum-Magnesium coatings such as ZM115 provide meaningfully better protection at cut edges than standard galvanizing: the magnesium and aluminum in the coating promote the formation of dense, adherent corrosion products, combined with galvanic (sacrificial) protection from the surrounding coating, which together slow corrosion at the exposed edge significantly compared to zinc-only coatings. This reduces — though does not entirely eliminate — the need for secondary edge treatment on thin-gauge, laser-cut components.

This benefit applies to the kerf-fold body components of the foundation, which retain the full ZM115 coating; the separately galvanized adapter plates receive their corrosion protection through the post-fabrication hot-dip process instead.

Technical Lifespan Underground (2 mm / 14-Gauge Steel)

The 2 mm material thickness provides a substantial structural safety margin. Technical lifespan — defined as the time until the material loses approximately 50% of its original thickness — depends heavily on the surrounding backfill:

In clean crushed stone / coarse gravel (well-drained): estimated technical lifespan of 75 to 80 years. Free-draining backfill keeps the coating in a low-corrosivity environment close to atmospheric conditions; the ZM115 coating is estimated to provide primary protection for approximately the first 25 years before the base steel begins to show measurable loss.

In compacted aggregate base course (0-32, moisture-retaining): estimated technical lifespan of 45 to 50 years. The denser, more moisture-retentive environment increases corrosion rates relative to free-draining gravel, but the magnesium-rich coating still substantially outperforms standard galvanizing in this environment.

These figures are engineering estimates derived from published corrosion-rate data for Zn-Al-Mg coatings and hot-dip galvanized steel in soil (approximately 1–2 $\mu\text{m}/\text{year}$ coating loss, and 10–35 $\mu\text{m}/\text{year}$ base steel loss once exposed, depending on soil drainage and moisture retention) — not laboratory results specific to this product. Actual service life depends on local soil resistivity, pH, and moisture content; a site-specific soil corrosivity assessment (e.g., per DIN 50929) is recommended for critical installations.

Sustainability & Recyclability

Our foundations support green building initiatives and circular economy principles. Manufactured in facilities operating under ISO 14001 environmental management standards, the coating alloy is free of volatile organic compounds (VOCs) and does not leach regulated heavy metals such as lead or cadmium into the surrounding soil. At the end of its service life, the foundation is fully recyclable through standard steel recycling streams without requiring de-coating or chemical separation, reducing its lifecycle carbon footprint relative to synthetic or composite alternatives.

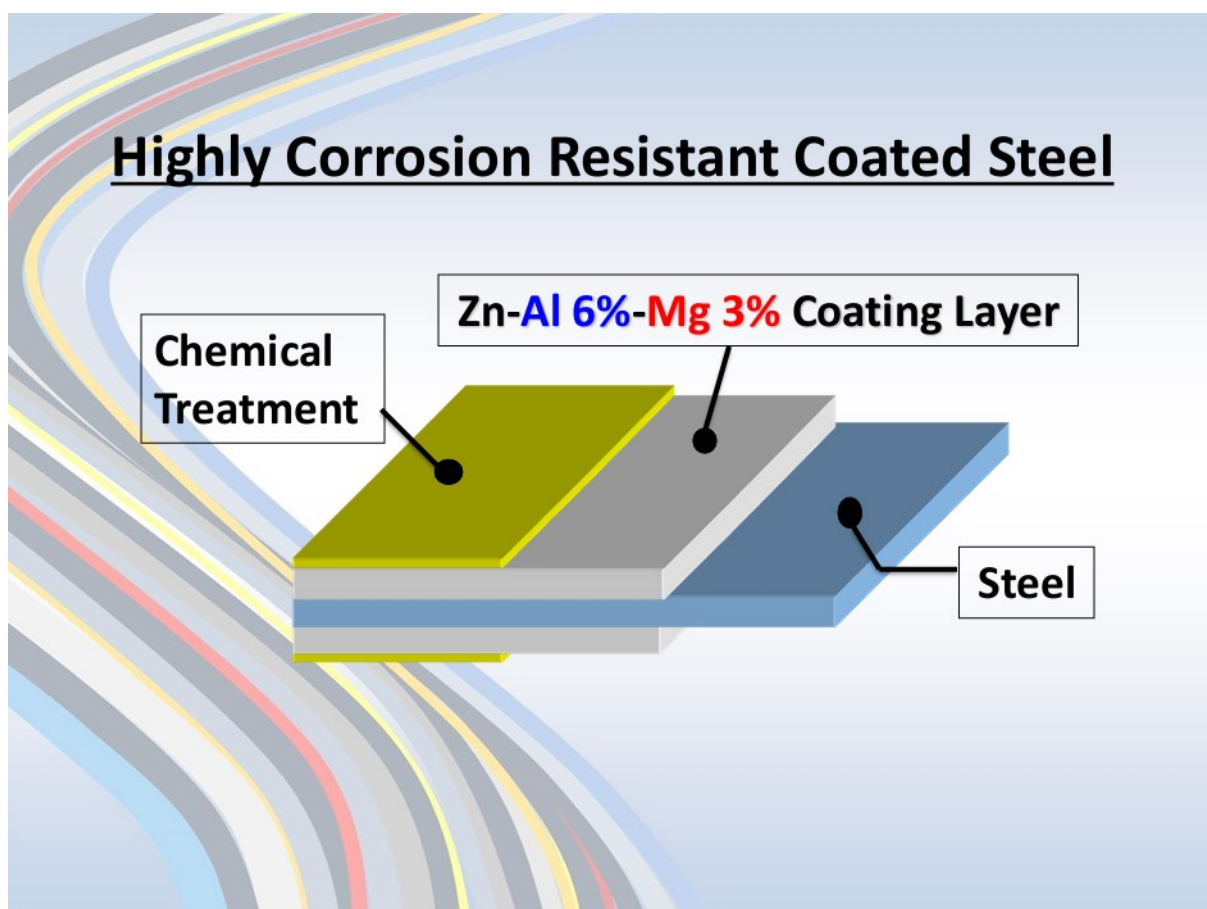


Image of ZAM layers on steel